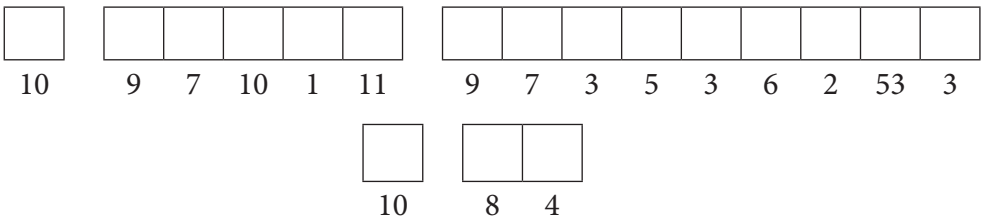
nelsonnet

WORKSHEET

Chapter 3: Optimisation

Cogito, ergo sum: …



René Descartes – philosopher

Questions

1 Find two numbers that add to 80 and whose product is a maximum.

2 Find two numbers whose difference is 32 and whose product is a minimum.

3 Find two numbers whose product is 36 and whose sum is a maximum.

4 Find two numbers whose product is 36 and whose sum is a minimum.

5 Find two numbers whose sum is -20 and whose product is a maximum.

6 A ball is catapulted upwards and follows a path of $f(x) = -\frac{1}{2}x^2 + 9x + 2$, where x is in seconds. At what time does it reach its maximum height after being thrown?

7 What is the ball's maximum height in metres?

8 A car is taken for a 5-minute drive. Its speed in km/h is given by $f(x) = x^3 - 27x^2 + 93x$, where x is the number of minutes. At what time does it reach its maximum speed?

9 What is the car's maximum speed during the drive?

10 The profit made by a company that manufactures x items per day is given by $P(x) = 20x^3 + 7.5x^2 - 6150x + 45\,000$. How many items should be made each day to maximise profit?

11 What is the maximum profit that can be made each day?

Solutions

A 2 min	**F** 9 s	**I** 10	**M** 6 and 6	**O** 16 and -16	**T** 86 km/h
E 1 and 36	**H** 42.5 m	**K** \$4250	**N** 40 and 40	**R** -10 and -10	

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